Popper, 1934: conjecture and refutation

Plate not present

`img/popper-lse.jpg`

See `img/IMAGE-SOURCES.txt`

Karl Popper, about 1980.

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Karl Popper, *The Logic of Scientific Discovery*, chapter 10, section 85 (1959), the authorised English of *Logik der Forschung* (1934).

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We do not know: we can only guess.

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Popper's tetradic schema

The way in which knowledge progresses … is by unjustified (and unjustifiable) anticipations, by guesses, by tentative solutions to our problems, by conjectures. These conjectures are controlled by criticism; that is, by attempted refutations …

Schema after Karl Popper, *Objective Knowledge*, chapter 8 (1972). Quotation: *Conjectures and Refutations*, preface (1963). English original.

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Popper's tetradic schema

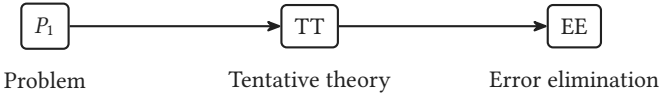
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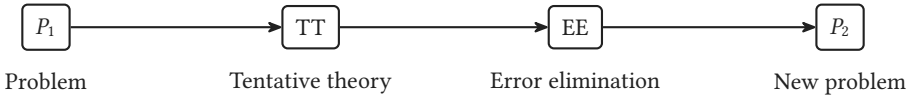
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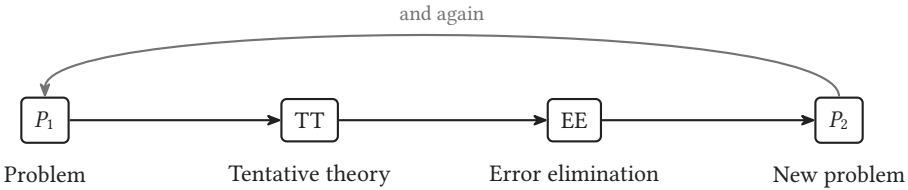
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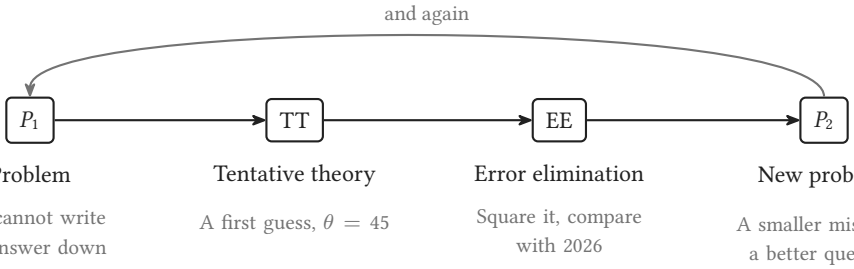
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Turing, 1950: the child machine



Alan Turing, 29 March 1951.

Elliott and Fry

A. M. Turing, “Computing Machinery and Intelligence”, *Mind* LIX(236), 1950, section 7, Learning Machines.

English original. The muted line is this deck’s paraphrase, not a translation.

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We cannot expect to find a good child-machine at the first attempt. One must experiment with teaching one such machine and see how well it learns. One can then try another and see if it is better or worse.

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Now the learning process may be regarded as a search for a form of behaviour which will satisfy the teacher (or some other criterion).

You do not get it right the first time. You try, you measure, you try again.

A. M. Turing, "Computing Machinery and Intelligence", *Mind* LIX(236), 1950, section 7, Learning Machines.

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Whewell, 1858: trying wrong guesses



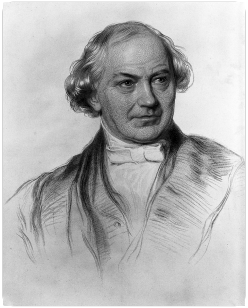
William Whewell, stipple engraving.

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William Whewell, *Novum Organon Renovatum*, third edition (1858), book II, chapter V, section I, page 79.

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Whewell, 1858: trying wrong guesses



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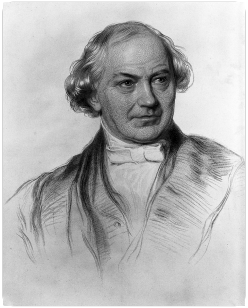
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To try wrong guesses is, with most persons, the only way to hit upon right ones.

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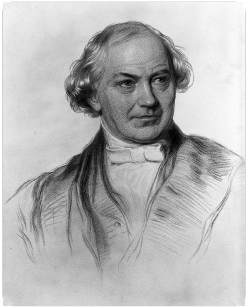
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To try wrong guesses is, with most persons, the only way to hit upon right ones.

The tendencies of our speculative nature … perpetually show their vigour by overshooting the mark. They obtain something, by aiming at much more.

Most guesses are wrong, and a step that aims too far is how the right one gets found.

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